

Chokro (চক্রে) — 16-Feature Implementation & Evaluation Guide

Complete technical reference across Module 1 (Lab 5), Module 2 (Lab 6), and Module 3 (Lab 7)

REPOSITORY sharzilnfz/chokro (Sprint3)	FEATURE QUOTA 4 Members × 4 Features = 16	TYPESCRIPT STATUS ✓ 0 Errors (4 packages)	TEST SUITE STATUS ✓ 40/40 Suites (332 Tests)
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1. Executive Summary & Evaluation Feature Matrix

Per the project guidelines, shared baseline infrastructure (Sign-in/OAuth, Sign-up, Logout, Role/Profile Management, and Generic Admin CRUD) is excluded from feature scores. Every team member has implemented 4 distinct, load-bearing full-stack features backed by database tables, domain service logic, route handlers, mobile/admin UI surfaces, and real external API integrations with graceful fallbacks.

FIGURE 1: FEATURE ALLOCATION BY TEAM MEMBER & EVALUATION MODULE

Member 1: Sadat SKD @SadatSKD · Trust & Settlement MOD 1 (LAB 5) · F1 Partner KYC OCR Gate API: Google Cloud Vision OCR MOD 2 (LAB 6) · F2 Trust Gate Fraud Engine API: Telegram Bot Webhooks MOD 3 (LAB 7) · F3 Escrow Settlement & Disputes API: Resend Email Inbound MOD 3 (LAB 7) · F4 Custody Handover OTP API: Twilio SMS Gateway	Member 2: Ahmad Sameer @Antizer17 · Marketplace & Discovery MOD 1 (LAB 5) · F1 Media EXIF Privacy Strip API: Cloudinary CDN Pipeline MOD 2 (LAB 6) · F2 Hyperlocal Radius Feed API: OSM Nominatim Geocoding MOD 3 (LAB 7) · F3 Reverse Demand Board API: OneSignal Push Notifications MOD 3 (LAB 7) · F4 Binding Counter-Offers API: Pusher / Ably Realtime	Member 3: Sharzil Nafis @sharzilnfz · Valuation & Logistics MOD 1 (LAB 5) · F1 Valuation Rate Cards API: Metals-API Commodities MOD 2 (LAB 6) · F2 AI Scrap Vision Agent API: OpenAI GPT-4o-mini MOD 3 (LAB 7) · F3 Smart Geo-Dispatch Routing API: Mapbox Matrix TSP MOD 3 (LAB 7) · F4 B2B Live Auction Engine API: Pusher Realtime Ticker	Member 4: Imran Ahmed Upom @Imran-1815 · Drop Zones & Impact MOD 1 (LAB 5) · F1 Zone Telemetry & Posters API: Google Static Maps MOD 2 (LAB 6) · F2 QR Deposit & Scale Gate API: Firebase Cloud Messaging MOD 3 (LAB 7) · F3 Green Wallet Cash-Out API: SSLCommerz MFS Gateway MOD 3 (LAB 7) · F4 Carbon Avoidance Ledger API: Climatiq ISO 14044 API
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2. End-to-End Product Lifecycle Architecture

FIGURE 2: UNIFIED CIRCULAR ECONOMY NARRATIVE FLOW

1. Capture & Valuation - Photo taken with camera m3 F2: AI Vision classifies item & condition m3 F1: Joined to benchmark rate card	2. Material Handover Campus Bin: m4 F2 QR deposit session + m4 F1 poster Pickup: m3 F3 Geo-dispatch + m1 F4 OTP custody transfer Marketplace: m2 F2 Geo-feed + m2 F3/F4 demand & bargaining	3. Trust Gate Verification m1 F2: Multi-signal fraud engine (pHash, scale divergence, velocity) - Clear \$\to\$ PENDING flips to VERIFIED - Flagged \$\to\$ Escalated to m1 F1/F2 admin queue	4. Financial & ESG Settlement m4 F3: Green Wallet MFS cash-out via SSLCommerz m4 F4: ESG certificate generation m3 F4 / m1 F3: B2B bulk live auctions & escrow release
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3. Code-Grounded Feature Implementation Details

Member 1 (m1) — Sadat SKD (@SadatSKD)		Domain: Trust Gate, Identity & Settlement
F1 (Module 1 / Lab 5): Partner KYC & Licence Document Intelligence		API: Google Cloud Vision OCR
Automated optical intelligence engine analyzing DoE trade licenses, extracting registration numbers and expiration dates, calculating normalized Levenshtein similarity, and enforcing e-waste authorization flags.		
Code Location: <code>apps/api/lib/domain/PartnerKycDomain.ts</code> , <code>apps/api/lib/repos/kycExtractions.ts</code>		DB Tables: <code>kyc_extractions</code> , <code>partner_compliance_audits</code> , <code>partners</code>
Endpoints: <code>POST /api/v1/partners/kyc/extract</code> , <code>GET /api/v1/admin/partners/kyc/queue</code>		UI Surfaces: <code>PartnerConsoleScreen.tsx</code> (M18), <code>/admin/kyc-queue</code> (A06)
Rubric Defense: Defends statutory DoE compliance under Bangladesh E-Waste Management Rules 2021 rather than a generic admin approve button.		
F2 (Module 2 / Lab 6): Trust Gate — Deposit Verification & Fraud Engine		API: Telegram Bot Webhooks
Multi-signal fraud adjudication engine processing perceptual image hashes (pHash), weight divergence ratios, account velocity, and geofences, clearing credits or alerting staff via Telegram.		
Code Location: <code>apps/api/lib/domain/TrustGateDomain.ts</code> , <code>CreditVerificationDomain.ts</code>		DB Tables: <code>trust_decisions</code> , <code>fraud_flags</code> , <code>trust_threshold_configs</code>
Endpoints: <code>POST /api/v1/trust-gate/evaluate</code> , <code>GET /api/v1/admin/trust-gate/escalations</code>		UI Surfaces: <code>/admin/trust-gate</code> (A07), <code>/admin/thresholds</code> (A08)
Rubric Defense: Mathematical fraud score stack and perceptual image deduplication rather than subjective manual review.		
F3 (Module 3 / Lab 7): Escrow Settlement & Quality Dispute Arbitration		API: Resend Email Inbound
Bilateral escrow holding mechanism for B2B scrap lots with countdown inspection timeouts, two-sided photo evidence dispute ticketing, and administrator arbitration.		
Code Location: <code>apps/api/lib/domain/EscrowDomain.ts</code> , <code>apps/api/lib/domain/DisputeDomain.ts</code>		DB Tables: <code>escrow_holds</code> , <code>disputes</code>
Endpoints: <code>POST /api/v1/disputes</code> , <code>POST /api/v1/admin/disputes/[id]/resolve</code>		UI Surfaces: <code>DisputeScreen.tsx</code> (M15), <code>/admin/disputes</code> (A09)
Rubric Defense: Manages bilateral financial liabilities and automatic fund release timeouts rather than a generic ticketing system.		
F4 (Module 3 / Lab 7): Chain-of-Custody Handover & OTP Handshake		API: Twilio SMS Gateway
Cryptographic 2-party handshake protocol where giver receives an SMS OTP verified by collector app to execute immutable physical custody transfer.		
Code Location: <code>apps/api/lib/domain/CustodyDomain.ts</code> , <code>HandoverDomain.ts</code>		DB Tables: <code>custody_handovers</code> , <code>pickup_orders</code>
Endpoints: <code>POST /api/v1/handovers/generate-otp</code> , <code>POST /api/v1/handovers/verify-otp</code>		UI Surfaces: <code>HandoverOtpModal.tsx</code> (M08), <code>PickupScreen.tsx</code> (M07)
Rubric Defense: Custody transfer is a signed cryptographic transaction that neither party can counterfeit alone.		

F1 (Module 1 / Lab 5): Listing Media Pipeline & EXIF Privacy Stripping

API: Cloudinary CDN Pipeline

Server-side multi-format media pipeline that parses uploads, extracts and removes residential GPS EXIF coordinates before storage, and generates responsive WebP images.

Code Location: `apps/api/lib/domain/MediaDomain.ts`,
`apps/api/lib/repos/listingMedia.ts`

DB Tables: `listing_media`, `listings`

Endpoints: `POST /api/v1/media/upload`, `POST /api/v1/listings`

UI Surfaces: `CreateListingScreen.tsx` (M02)

Rubric Defense: Enforces residential privacy at the ingest seam rather than raw client-side file uploads.

F2 (Module 2 / Lab 6): Hyperlocal Geo-Discovery & Radius Filter Feed

API: OSM Nominatim Geocoding

Location-aware feed computing trigonometric SQL Haversine distances in PGLite WASM and PostgreSQL, filtering listings by radius (1–10 km) and Bangladesh administrative Thana/Ward units.

Code Location: `apps/api/lib/domain/FeedDomain.ts`,
`apps/api/lib/repos/feed.ts`

DB Tables: `listings` (with `lat`, `lng`,
`thana`), `service_areas`

Endpoints: `GET /api/v1/feed`, `GET /api/v1/listings/feed`

UI Surfaces: `FeedScreen.tsx` (M01)

Rubric Defense: Solves localized transport economics with spatial SQL ranking instead of static nationwide lists.

F3 (Module 3 / Lab 7): Reverse Marketplace — Recycler Demand Board & Matching

API: OneSignal Push Notifications

Inverted marketplace board where commercial recyclers post standing material demands; triggers automated spatial push matching when new seller listings match.

Code Location: `apps/api/lib/domain/DemandDomain.ts`,
`DemandBoardDomain.ts`

DB Tables: `buyer_demands`, `demand_matches`

Endpoints: `POST /api/v1/demands`, `GET /api/v1/demands/matches`

UI Surfaces: `DemandsScreen.tsx` (M11)

Rubric Defense: Algorithmic event-driven matching engine that solves circular supply chain fragmentation.

F4 (Module 3 / Lab 7): Binding Counter-Offer Negotiation Engine

API: Pusher / Ably Realtime

Bilateral contract state machine supporting formal offer/counter-offer cycles, 24-hour TTL expiry, and atomic locking of listings upon binding acceptance.

Code Location: `apps/api/lib/domain/NegotiationDomain.ts`,
`apps/api/lib/repos/negotiations.ts`

DB Tables: `negotiation_threads`,
`negotiation_offers`

Endpoints: `POST /api/v1/negotiations`, `POST`
`/api/v1/negotiations/[id]/offer`

UI Surfaces: `NegotiationThreadScreen.tsx`
(M12)

Rubric Defense: Replaces informal chat with enforceable contract state machines and race-condition-safe locks.

F1 (Module 1 / Lab 5): Market-Benchmarked Valuation & Rate Cards

API: Metals-API / AlphaVantage

Dynamic pricing engine benchmarking local BDT scrap prices against international commodity spot feeds with FX conversion, drift alerts, and effective-dated rate supersession.

Code Location: `apps/api/lib/domain/RateCardDomain.ts`,
`ValuationDomain.ts`

DB Tables: `rate_card_entries`,
`rate_benchmarks`

Endpoints: `GET /api/v1/rate-card/published`, `GET /api/v1/rate-card/estimate`

UI Surfaces: `RateCardScreen.tsx` (M04),
`/admin/rate-card` (A04)

Rubric Defense: Rate cards use immutable effective-dated supersession; past valuations stay 100% reproducible.

F2 (Module 2 / Lab 6): AI Next-Life Scrap Vision Agent

API: OpenAI GPT-4o-mini Vision

Multi-modal scrap classification agent determining material category, condition band, estimated weight/pieces, and circular path, with mandatory hazardous e-waste safety locks.

Code Location: `apps/api/lib/domain/VisionDomain.ts`,
`ValuationDomain.ts`

DB Tables: `valuation_scans`

Endpoints: `POST /api/v1/valuation/classify-and-estimate`

UI Surfaces: `VisionScanScreen.tsx` (M03)

Rubric Defense: Constrains multi-modal AI output to domain enums and joins directly to effective database rate cards.

F3 (Module 3 / Lab 7): Smart Geo-Dispatch & Route Optimizer

API: Mapbox Matrix TSP API

Capacity-constrained collector dispatching and Traveling Salesperson Problem (TSP) multi-stop route sequencing based on vehicle payload limits, distance matrices, and partner category licenses.

Code Location: `apps/api/lib/domain/DispatchDomain.ts`,
`PickupDomain.ts`

DB Tables: `pickup_orders`,
`dispatch_assignments`

Endpoints: `POST /api/v1/pickups/book`, `GET /api/v1/pickups/tasks`

UI Surfaces: `PickupScreen.tsx` (M07)

Rubric Defense: Solves vehicle payload allocation and sequential stop ordering rather than rendering passive map markers.

F4 (Module 3 / Lab 7): B2B Bulk Scrap Live Auction Engine

API: Pusher Channels Realtime

Real-time auction engine for institutional bulk lots featuring server-authoritative monotonic bid ordering, sealed reserve price protection, and dynamic anti-snipe countdown extensions.

Code Location: `apps/api/lib/domain/AuctionDomain.ts`,
`apps/api/lib/repos/auctions.ts`

DB Tables: `auction_lots`, `auction_bids`

Endpoints: `GET /api/v1/auction-lots/live`, `POST /api/v1/auction-lots/[id]/bids`

UI Surfaces: `AuctionsScreen.tsx` (M09),
`LotDetailScreen.tsx` (M10)

Rubric Defense: Race-condition-free bidding engine with database transaction locks and server-authoritative ordering.

F1 (Module 1 / Lab 5): Drop-Zone Telemetry & Printable Poster Generator

API: Google Static Maps

Physical collection bin capacity modeling with predictive fill rate alerts and a printable QR poster generator featuring HMAC-SHA256 tamper-evident tokens and static maps.

Code Location: `apps/api/lib/domain/DropZoneTelemetryDomain.ts`, `apps/api/lib/qr.ts` DB Tables: `drop_zones`, `zone_capacity_logs`

Endpoints: `GET /api/v1/drop-zones`, `GET /api/v1/drop-zones/[id]/poster` UI Surfaces: `/admin/drop-zones` (A02), `/admin/zone-capacity` (A03)

Rubric Defense: Physical-digital bridge using timing-safe cryptographic QR verification and predictive fill models.

F2 (Module 2 / Lab 6): Smart QR Deposit Session & Weight-Scale Gate

API: Firebase Cloud Messaging

Time-bounded deposit sessions (15m TTL) requiring users to scan bin QRs, upload camera-only evidence, and reconcile declared weight with collector scale telemetry upon emptying.

Code Location: `apps/api/lib/domain/DepositDomain.ts`, `ZoneEmptyingDomain.ts` DB Tables: `drop_sessions`, `deposit_records`, `zone_emptying_records`

Endpoints: `POST /api/v1/drop-sessions`, `POST /api/v1/deposits` UI Surfaces: `QRScannerScreen.tsx` (M05), `DepositFlowScreen.tsx` (M06)

Rubric Defense: Mints credits strictly inside hardware-validated physical sessions to prevent remote fraudulent claims.

F3 (Module 3 / Lab 7): Green Wallet Payout & MFS Cash-Out Engine

API: SSLCommerz MFS Gateway

Append-only transaction ledger with balance classifiers, concurrency locks, monthly liability budget caps, and mobile financial services (bKash/Nagad via SSLCommerz) automated cash-outs with compensating reversal entries.

Code Location: `apps/api/lib/domain/SettlementDomain.ts`, `WalletDomain.ts`, `LedgerMath.ts` DB Tables: `credit_txns`, `redemption_requests`, `payout_records`

Endpoints: `GET /api/v1/wallet/balance`, `POST /api/v1/wallet/redemptions` UI Surfaces: `WalletScreen.tsx` (M13), `RedemptionRequestScreen.tsx` (M14), `/admin/redemptions` (A10)

Rubric Defense: Balances are never directly mutated; cash-outs execute as 2-phase transactions with compensating reversal refs.

F4 (Module 3 / Lab 7): Carbon Avoidance Ledger & ESG Certificate Generator

API: Climatiq ISO 14044 API

Immutable corporate ESG impact ledger translating verified physical scrap weights into avoided CO2e using ISO 14044 emission factors, generating SHA-256 verifiable sustainability certificates.

Code Location: `apps/api/lib/domain/ImpactDomain.ts`, `apps/api/lib/repos/impact.ts` DB Tables: `impact_records`, `emission_factors`, `sustainability_certificates`

Endpoints: `GET /api/v1/impact/personal`, `GET /api/v1/certificates/[ref]` UI Surfaces: `ImpactDashboardScreen.tsx` (M16), `CertificateViewScreen.tsx` (M17), `/admin/certificates` (A12)

Rubric Defense: Generates legally auditable corporate compliance documents backed by immutable SHA-256 cryptographic hashes.

4. Complete Screen Directory & Architectural Invariants

Mobile Screens ([apps/mobile/src/screens](#))

Admin Console Pages ([apps/api/app/admin](#))

ID	PAGE ROUTE	OWNER	FUNCTION
A01	campuses/page.tsx	Core	Campus directory & registry
A02	drop-zones/page.tsx	m4	Bin QR configuration
A03	zone-capacity/page.tsx	m4	Telemetry fill-rate monitor
A04	rate-card/page.tsx	m3	Effective rate versioning
A05	partners/page.tsx	m1	Partner directory & fleet
A06	kyc-queue/page.tsx	m1	Side-by-side OCR doc view
A07	trust-gate/page.tsx	m1	Escalation worklist
A08	thresholds/page.tsx	m1	Fraud score threshold config
A09	disputes/page.tsx	m1	Escrow arbitration desk
A10	redemptions/page.tsx	m4	MFS cash-out queue
A11	liability/page.tsx	m4	Monthly platform pool metrics
A12	certificates/page.tsx	m4	ESG certificate issuer

ID	SCREEN FILE	OWNER	FUNCTION
M01	FeedScreen.tsx	m2	1–10 km radius slider & feed
M02	CreateListingScreen.tsx	m2	EXIF stripped photo uploader
M03	VisionScanScreen.tsx	m3	GPT-4o vision scanner
M04	RateCardScreen.tsx	m3	Benchmark rates & drift info
M05	QRScannerScreen.tsx	m4	Signed bin QR viewfinder
M06	DepositFlowScreen.tsx	m4	15m deposit session flow
M07	PickupScreen.tsx	m3	Collector TSP route & tasks
M08	HandoverOtpModal.tsx	m1	6-digit custody OTP modal
M09	AuctionsScreen.tsx	m3	Live B2B scrap lots
M10	LotDetailScreen.tsx	m3	Real-time bidding desk
M11	DemandsScreen.tsx	m2	Demand composer & matches
M12	NegotiationThreadScreen.tsx	m2	24h TTL counter-offer desk
M13	WalletScreen.tsx	m4	Verified vs Pending wallet
M14	RedemptionRequestScreen.tsx	m4	MFS cash-out interface
M15	DisputeScreen.tsx	m1	Bilateral dispute ticket filer
M16	ImpactDashboardScreen.tsx	m4	Avoided CO2e metrics
M17	CertificateViewScreen.tsx	m4	Verifiable ESG

ID	SCREEN FILE	OWNER	FUNCTION
			certificate
M18	PartnerConsoleScreen.tsx	m1	DoE license KYC & fleet

5. Architectural Invariants & CLI Commands

1. Measurement Invariant: `APPLIANCES` , `E_WASTE` `$\to$` `unit: 'piece'` (integer `piece_count`). All other 7 categories `$\to$` `unit: 'kg'` (decimal `declared_weight`). Enforced by Zod superRefine in `@chokro/shared` .

2. Immutable Ledger Math: Balances are never updated in place. Calculated via `apps/api/lib/LedgerMath.ts` as `$\sum` `\text{amount}` where `status = 'VERIFIED'` . Flips from Pending to Verified are owned strictly by `CreditVerificationDomain.ts` .

3. Seam Isolation: Route handlers are thin controllers wrapped in `safeRoute()` . Business domains in `apps/api/lib/domain/` never import `@chokro/db` . Repositories wrap all queries in `withDb()` transactions.